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Cumulative Adverse Childhood Experiences and Parent-Reported Allergic Conditions and Asthma Among U.S. Children: A Nationally Representative Study

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Allergic diseases are common chronic conditions in childhood, yet the role of adverse childhood experiences (ACEs) in their development remains incompletely understood at the population level. Using pooled data from the 2016–2023 National Survey of Children's Health, a nationally representative survey of U.S. children aged 0–17 years, we examined associations between ACEs and parent-reported allergic conditions and asthma, including parent-reported disease severity (moderate-to-severe). ACEs were assessed across nine parent-reported domains and summarized as a cumulative score. Weighted logistic regression and restricted cubic spline models were performed to estimate associations between ACEs and allergic outcomes, with subgroup analyses by age and sociodemographic characteristics. A total of 290,599 children were included (*M* [*SD*] age = 8.62 [5.16] years; 51.17% male), representing 61.5 million children. The weighted prevalence was 20.09% for allergic conditions and 7.18% for asthma. Compared with no ACEs, exposure to ≥4 ACEs was associated with higher odds of parent-reported allergic conditions (*OR* = 1.85; 95% CI [1.68, 2.04]) and asthma (*OR* = 2.08; 95% CI [1.79, 2.41]). Associations were strongest in early childhood. Among children aged 0–2 years, those exposed to ≥4 ACEs had 2.6-fold higher odds of allergic conditions (*OR* = 2.64; 95% CI [1.65, 4.23]) and nearly fivefold increased odds of asthma (*OR* = 4.71; 95% CI [2.24, 9.91]), compared to those with no ACEs. In item-level analyses, caregiver mental illness and experiences of racial discrimination showed the strongest associations with allergic outcomes. These findings highlight early-life adversity as an important social determinant of allergic disease and the need for early identification and prevention strategies.

Public Significance Statement

This nationally representative study found that children exposed to more adverse childhood experiences had higher odds of parent-reported allergic conditions and asthma, with the strongest associations observed in the youngest children. These findings suggest that early adversity may be an important social factor in childhood allergic health and support early identification and trauma-informed approaches in pediatric allergy and asthma care.

Keywords: adverse childhood experiences, pediatric allergic diseases, asthma, psychosocial stress, social determinants of health

Supplemental materials: https://doi.org/10.1037/amp0001740.supp

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The National Survey of Children's Health survey protocol was reviewed and approved by the U.S. Census Bureau's Disclosure Review Board. All data are de-identified and publicly available, and informed consent was obtained from parents or guardians by the Census Bureau. More details are available at https://www.childhealthdata.org/learn-about-the-nsch/methods. The authors have declared that no competing interests exist.

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continued

Adverse childhood experiences (ACEs) are commonly conceptualized as early-life exposures to chronic or recurrent social and familial adversities that shape developmental trajectories of health across the life course (Bhutta et al., 2023). Rather than operating through single, linear pathways, ACEs are increasingly understood to influence mental and physical health through complex, interacting processes that unfold over time and are shaped by resilience and adaptation (Hays-Grudo et al., 2021). Nearly 60% of children worldwide have experienced at least one ACE, with over one third exposed to multiple adversities (Madigan et al., 2025). Therefore, understanding how ACEs relate to diverse health outcomes is essential for clarifying the developmental roots of lifelong risk.

ACEs and Lifespan Health Outcomes

First proposed by Felitti and colleagues in 1998 (Felitti et al., 1998), ACEs refer to various domains of potentially traumatic events occurring in early life, offering a cumulative and quantifiable framework for assessing their long-term health consequences. Subsequent decades of research have established robust links between ACE exposure and a wide spectrum of mental health conditions, including depression, anxiety, and suicidality (Daníelsdóttir et al., 2024; Weems et al., 2021). Over time, growing evidence has extended these associations to physical health outcomes, such as obesity, hypertension, and cardiovascular disease (Bhutta et al., 2023; Moriya et al., 2024). However, the relationship between ACEs and health outcomes is far from deterministic, and considerable uncertainty remains regarding how, when, and in whom early adversity translates into specific health risks.

Contemporary life-course models, such as the Intergenerational and Cumulative Adverse and Resilient Experiences model proposed by Hays-Grudo et al. (2021), emphasize the dynamic interplay among adversity, adaptation, and resilience in shaping long-term health across critical developmental windows, from the prenatal period through childhood. Consistent with this perspective, ACE exposure does not produce uniform outcomes; rather, the health consequences of similar adversity profiles may diverge depending on developmental timing and the availability of buffering supports. Notably, the immune system is highly plastic during the prenatal and early postnatal periods, rendering immune-mediated outcomes particularly sensitive to early-life adversity (Beyer et al., 2025). Maternal experiences of anxiety, depression, major negative life events, or exposure to violence during pregnancy have been linked to higher risks of wheeze and asthma in offspring (Adgent et al., 2024; Rosa et al., 2018). These features underscore the value of allergic diseases—including, but not limited to, asthma—as meaningful outcomes for understanding the developmental consequences of early adversity from the prenatal period through childhood.

Allergic Diseases and Asthma in the Context of ACEs

Children with allergic diseases, including allergic rhinitis, asthma, atopic dermatitis, and food allergies, represent a large portion of this burden (Pawankar, 2014; Shin et al., 2023). These conditions are among the most prevalent chronic diseases in childhood, affecting approximately 10%–30% of children globally, and their prevalence has been rising in recent decades (Koo et al., 2023; Pate et al., 2022; Shin et al., 2023). In the United States, national data suggest that approximately one in five children are now affected, reflecting a continued rise in parent-reported allergic conditions (Pate et al., 2022). Beyond symptom burden, pediatric allergic diseases are linked to missed school days, impaired daily functioning, and reduced quality of life, and they often cluster developmentally along the “allergic march” (Dharmage et al., 2022; Ma et al., 2024), underscoring the importance of identifying early-life determinants that shape both onset and severity. Allergic diseases have also been linked to elevated risks of depression, anxiety, sleep disturbance, and suicidality, including among individuals with atopic dermatitis (Patel et al., 2019) and allergic rhinitis (Rodrigues et al., 2021).

Before the ACE framework was widely adopted, research had already shown that social and psychological stressors may shape allergic disease risk. Children in low-income households are more likely to live in substandard housing and be exposed to indoor allergens such as mold, pests, and

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environmental tobacco smoke, which can trigger allergic sensitization and respiratory inflammation (Perry et al., 2024). Moreover, perceived discrimination has been associated with both higher asthma prevalence and poorer asthma control among adolescents from racial and ethnic minority backgrounds (Thakur et al., 2017). Although not always explicitly framed as “ACEs,” these experiences reflect key domains of early adversity. Growing evidence also supports the role of prenatal adversity. Maternal depression, anxiety, and exposure to violence during pregnancy have also been associated with elevated risks of wheeze and asthma in offspring (Rosa et al., 2018). Adgent et al. (2024) followed 2,056 mother–child dyads from pregnancy through age 6 and found that a greater number of maternal stressful life events during pregnancy predicted a higher risk of wheeze at ages 4–6, particularly among boys. Likewise, a prospective birth cohort study of 3,387 families found that parental ACE exposure was associated with increased risk of allergic disease in children at age 5 (Freeman et al., 2024).

Taken together, existing literature suggests intriguing links between early-life stress and allergic disease, with particular attention to asthma and prenatal or maternal adversity. However, limited evidence is available on whether ACEs occurring during postnatal development are associated with allergic outcomes—particularly broader allergic conditions beyond asthma—in large, population-based pediatric samples. Given the high prevalence of pediatric allergic diseases and the importance of early-life environments for immune development, childhood ACE exposure may also be relevant to pediatric allergic conditions and asthma. We therefore hypothesized that ACEs experienced during childhood would be associated with higher odds of parent-reported allergic conditions and asthma and that greater cumulative ACE exposure would exhibit a graded association with these outcomes.

The Present Study

In this study, we pooled data from the 2016–2023 cycles of the National Survey of Children’s Health (NSCH) to evaluate the associations between ACE exposure and pediatric allergic diseases, including parent-reported allergic conditions and asthma. Asthma was examined separately from allergic conditions because, although allergic inflammation contributes importantly to asthma pathogenesis, asthma is clinically heterogeneous and not uniformly allergic in origin, including nonallergic forms such as exercise-induced, virus-related, and premenstrual asthma (Klain et al., 2022). We characterized ACE exposure both cumulatively and at the level of individual ACE domains to examine whether associations differed across adversity types. We also examined dose–response patterns using restricted cubic splines (RCS). Allergic outcomes were assessed in terms of both disease presence and parent-reported disease severity. In addition, we assessed whether the observed associations were modified by key sociodemographic factors, including child age, sex, race/ethnicity, family poverty level (FPL), household smoking exposure, family structure, and region.

Method

Study Population

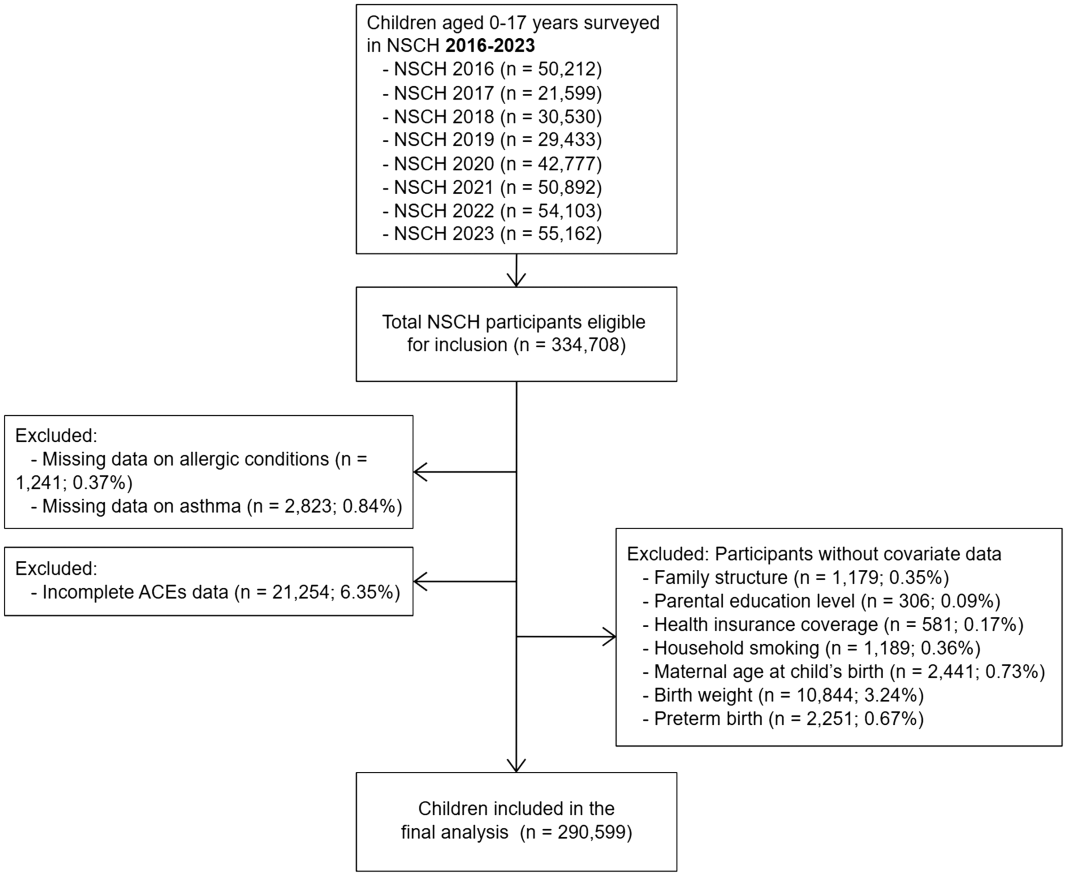
Following Census Bureau guidance (Bureau, 2024), this study analyzed pooled data from the 2016–2023 cycles of the NSCH, a nationally representative, cross-sectional survey of U.S. children aged 0–17 years. Conducted by the U.S. Census Bureau, the NSCH collects parent-reported information on children’s health, development, and family environment (Bureau, 2023, 2024, 2025; Ghandour et al., 2018). One child per household is randomly selected, and survey weights account for sampling design and nonresponse to ensure national representativeness. Since 2016, the NSCH has used a self-administered, mail-based format with updated questionnaires and weighting procedures (Ghandour et al., 2018). All data are de-identified and publicly available, and informed consent was obtained. As shown in Figure 1, children with missing information on allergic conditions, asthma, ACEs, or any key covariates were excluded. The final analytical sample consisted of 290,599 participants.

Allergic Conditions and Asthma

The primary outcomes were current allergic conditions, moderate-to-severe allergic conditions, current asthma, and moderate-to-severe asthma. Allergic condition was defined based on two NSCH parent-reported items: (a) “Has a doctor or other health care provider EVER told you that this child has allergies (such as including food, drug, insect, seasonal, or other)?” and (b) “If yes, does this child CURRENTLY have the condition?” Children whose parent or guardian answered “Yes” to both were classified as having a current allergic condition. Among children with current allergic conditions, parent-reported allergic conditions severity was assessed using the NSCH single item: “If yes, is it mild, moderate, or severe?”; responses of “moderate” or “severe” were classified as moderate-to-severe allergic conditions.

Asthma was defined using parallel criteria: (a) “Has a doctor or other health care provider EVER told you that this child has asthma?” and (b) “If yes, does this child CURRENTLY have the condition?” Current asthma was assigned when both responses were “Yes.” Among children with current asthma, parent-reported asthma severity was assessed using the parallel single item: “If yes, is it mild, moderate, or severe?”; responses of “moderate” or “severe” were classified as parent-reported moderate-to-severe asthma. Asthma was examined as a separate outcome from allergic conditions because it is clinically heterogeneous and not

Figure 1
Flowchart of Study Population Selection



Note. ACE = adverse childhood experience; NSCH = National Survey of Children's Health.

uniformly allergic in origin. These outcomes are not mutually exclusive and may co-occur.

ACEs

The primary exposure was ACEs, assessed in the NSCH through nine parent-reported items regarding the child's early-life experiences of adversity: (a) parental separation or divorce; (b) parental death; (c) parental incarceration; (d) witnessing household violence (parents or other adults slap, hit, kick, or punch one another); (e) being a victim of or witnessing violence in the neighborhood; (f) living with someone who was mentally ill, suicidal, or severely depressed; (g) living with someone who had alcohol or drug problems; (h) being treated or judged unfairly because of race or ethnicity; and (i) experiencing economic hardship, defined as the caregiver reporting it was "somewhat often" or "very often" hard to get by on the family's income for basic needs such as food or housing. Each ACE was coded dichotomously (*present* = 1, *absent* = 0), and a cumulative ACE score ranging from 0 to 9 was calculated. ACE exposure was analyzed both as a continuous variable and categorically (0, 1, 2, 3, or ≥4), consistent with prior NSCH-based studies (Mansuri et al., 2020; Walker et al., 2022).

Covariates

Covariates included demographic, socioeconomic, perinatal, and household factors previously associated with both ACE exposure and allergic outcomes. Demographic variables included child's age (continuous and categorized as 0–2, 3–5, 6–11, or 12–17 years), sex (male/female), and race/ethnicity (non-Hispanic White, non-Hispanic Black, Hispanic, and Other). Socioeconomic variables included FPL (as a percentage of the federal poverty level: <100%, 100%–199%, 200%–399%, ≥400%), region of residence (Northeast, Midwest, South, West), family structure (two-parent married, two-parent unmarried, single mother, other), and highest parental education level (<high school, high school graduate, some college or associate degree, or college degree or higher). Perinatal and household environment variables included maternal age at the child's birth (18–24, 25–29, 30–34,

35–39, ≥40 years), household smoking (yes vs. no), birth weight (<2,500 g vs. ≥2,500 g), and preterm birth (yes vs. no). Insurance coverage (insured/uninsured) and survey year (2016–2023) were also included.

Statistical Analysis

All analyses accounted for the NSCH's complex survey design, incorporating sampling weights, strata, and primary sampling units to yield nationally representative estimates. For pooled analyses, annual weights were divided by the number of survey years. Descriptive statistics included weighted means (standard deviations) for continuous variables and unweighted counts with weighted percentages for categorical variables. Group comparisons were performed using weighted t tests (continuous) and χ^2 tests (categorical), or the Kruskal–Wallis rank-sum test for multiple groups. Standardized mean differences were calculated to assess effect sizes. Time trends were visualized using bar charts with 95% confidence intervals (CIs). Multivariable logistic regression estimated odds ratios (*OR*s) and 95% CIs for the association between ACE scores and each allergic outcome. Four models were constructed: Model 1, unadjusted; Model 2, adjusted for age, sex, and race/ethnicity; Model 3, additionally adjusted for region, family structure, FPL, parental education, health insurance, and survey year; and Model 4, fully adjusted for maternal age at child's birth, household smoking, birth weight, and preterm birth. RCS models with four knots (ACE scores: 0, 2, 4, 6) assessed potential nonlinear dose–response associations. Subgroup analyses were conducted stratified by child age group, sex, race/ethnicity, FPL, household smoking exposure, region, and family structure. A two-sided p value < .05 was considered statistically significant.

Transparency and Openness

This study was performed and reported in accordance with the Strengthening the Reporting of Observational Studies in Epidemiology reporting guideline. We reported how we determined our sample size, all data exclusions, all measures, and all analyses in the study. The data analyzed in this study are publicly available from the NSCH repository. All statistical analyses were performed using R (Version 4.4.1), with the "survey" package used for complex survey analyses and "ggplot2" for figure generation. This study was not preregistered.

Results

Study Population and Baseline Characteristics

From the 2016–2023 NSCH cycles, a total of 290,599 children were included, representing an estimated 61,503,745 U.S. children aged 0–17 years after applying survey weights. Baseline characteristics before and after exclusions are presented in Supplemental Table 1. Overall, the analyzed sample was broadly similar to the full sample in demographic and health characteristics, with only modest differences in some socioeconomic indicators. The weighted prevalence was 20.09% for parent-reported allergic conditions and 7.18% for parent-reported asthma.

As shown in Table 1, children with allergic conditions were older than those without allergic conditions (M_{age} 10.02 vs. 8.27 years; $p < .001$), and children with asthma were older than those without asthma (M_{age} 10.52 vs. 8.48 years; $p < .001$). The distributions of sex, race/ethnicity, region, family structure, parental education, and FPL all differed significantly between children with and without allergic conditions, as well as between children with and without asthma (all design-based χ^2, $p < .001$). The distribution of ACE categories among all participants was as follows: 0 (60.60%), 1 (21.71%), 2 (8.60%), 3 (4.10%), and ≥4 (4.99%). Children with allergic conditions or asthma had a higher ACE burden than those without allergic conditions or asthma, respectively (both $p < .001$). The weighted prevalence of individual ACEs by allergic conditions and asthma status is provided in Supplemental Table 2, while baseline characteristics stratified by ACE score groups are summarized in Supplemental Table 3.

Time Trends of Allergic Conditions and Asthma Prevalence

As shown in Figure 2A, across 2016–2023, the weighted prevalence of parent-reported allergic conditions increased from 19.94% in 2016 to 23.28% in 2023 among U.S. children aged 0–17 years (p for trend < .001). Parent-reported moderate-to-severe allergic conditions changed little, from 8.60% to 8.99% (p for trend = 0.159). In contrast, Figure 2B showed that the weighted prevalence of parent-reported asthma declined from 8.48% in 2016 to 6.79% in 2023 among U.S. children overall (p for trend < .001). Additionally, Supplemental Figure 1 showed temporal patterns of cumulative ACE scores across survey years and the pooled prevalence of individual ACEs. Year-specific estimates used the NSCH annual weights, and pooled estimates used weights divided by the number of pooled years.

Associations Between ACEs and Parent-Reported Allergic Conditions and Asthma

Descriptive analyses showed a graded increase in parent-reported allergic conditions and asthma prevalence across ACE score groups (Supplemental Figure 2), and these patterns were further confirmed by logistic regression analyses (Table 2). Compared to children with no ACEs, those exposed to 1, 2, 3, and ≥4 ACEs showed progressively higher

Table 1

Baseline Characteristics of the Study Population Stratified by Allergic Conditions and Asthma Status

Characteristic	Overall	Allergic conditions: No allergy	Allergic conditions: Allergy	SMD	*p*	Asthma: No asthma	Asthma: Asthma	SMD	*p*
Population, no. [weighted no.] (%)	290,599 [61,503,745] (100.00)	224,951 [49,147,643] (79.91)	65,648 [12,356,102] (20.09)			269,900 [57,087,776] (92.82)	20,699 [4,415,969] (7.18)		
Age, mean (*SD*), year	8.62 (5.16)	8.27 (5.22)	10.02 (4.65)	0.343	<.001[a]	8.48 (5.19)	10.52 (4.38)	0.397	<.001[a]
Age group, *n* (%)					<.001[b]				<.001[b]
0–2	44,159 (15.90)	39,388 (18.07)	4,771 (7.27)	0.329		43,370 (16.82)	789 (4.10)	0.425	
3–5	54,578 (16.45)	44,965 (17.37)	9,613 (12.76)	0.129		51,917 (16.82)	2,661 (11.67)	0.148	
6–11	84,523 (33.57)	63,435 (32.58)	21,088 (37.48)	0.103		77,795 (33.20)	6,728 (38.31)	0.107	
12–17	107,339 (34.08)	77,163 (31.97)	30,176 (42.48)	0.219		96,818 (33.17)	10,521 (45.91)	0.263	
Sex, *n* (%)					<.001[b]				<.001[b]
Male	150,254 (51.17)	114,339 (50.44)	35,915 (54.06)	0.073		138,452 (50.72)	11,802 (56.90)	0.124	
Female	140,345 (48.83)	110,612 (49.56)	29,733 (45.94)	0.073		131,448 (49.28)	8,897 (43.10)	0.124	
Race/ethnicity, *n* (%)					<.001[b]				<.001[b]
Hispanic	36,749 (25.05)	29,166 (26.15)	7,583 (20.66)	0.130		33,812 (25.09)	2,937 (24.50)	0.014	
Non-Hispanic White	200,355 (52.14)	154,883 (51.62)	45,472 (54.22)	0.052		187,403 (52.83)	12,952 (43.17)	0.194	
Non-Hispanic Black	16,165 (11.93)	11,815 (11.37)	4,350 (14.14)	0.083		13,892 (11.11)	2,273 (22.48)	0.308	
Non-Hispanic Other	37,330 (10.89)	29,087 (10.86)	8,243 (10.99)	0.004		34,793 (10.97)	2,537 (9.86)	0.036	
Region, *n* (%)					<.001[b]				<.001[b]
Midwest	71,636 (21.42)	55,632 (21.38)	16,004 (21.59)	0.005		66,854 (21.47)	4,782 (20.76)	0.017	
Northeast	50,366 (15.83)	38,915 (15.75)	11,451 (16.18)	0.012		46,435 (15.71)	3,931 (17.46)	0.047	
South	86,984 (38.67)	65,749 (37.93)	21,235 (41.60)	0.075		80,562 (38.57)	6,422 (39.96)	0.028	
West	81,613 (24.08)	64,655 (24.95)	16,958 (20.63)	0.103		76,049 (24.25)	5,564 (21.83)	0.057	
Family structure, *n* (%)					<.001[b]				<.001[b]
Two parents, married	214,141 (67.76)	166,644 (68.14)	47,497 (66.25)	0.040		200,772 (68.65)	13,369 (56.24)	0.258	
Two parents, unmarried	17,714 (7.83)	14,061 (7.98)	3,653 (7.22)	0.029		16,272 (7.73)	1,442 (9.12)	0.050	
Single mother	39,886 (16.49)	29,215 (15.82)	10,671 (19.14)	0.087		35,544 (15.78)	4,342 (25.65)	0.245	
Other	18,858 (7.92)	15,031 (8.06)	3,827 (7.39)	0.025		17,312 (7.84)	1,546 (8.99)	0.041	
Parent educational level, *n* (%)					<.001[b]				<.001[b]
Less than high school	6,143 (7.97)	5,232 (8.75)	911 (4.86)	0.155		5,605 (7.90)	538 (8.83)	0.034	
High school	34,517 (18.18)	27,467 (18.55)	7,050 (16.73)	0.048		31,530 (17.95)	2,987 (21.17)	0.081	
Some college or associate degree	62,536 (20.90)	47,858 (20.48)	14,678 (22.61)	0.052		57,211 (20.53)	5,325 (25.69)	0.123	
College degree or higher	187,403 (52.95)	144,394 (52.23)	43,009 (55.80)	0.072		175,554 (53.61)	11,849 (44.32)	0.187	
FPL, *n* (%)					<.001[b]				<.001[b]
<100%	26,664 (15.54)	20,743 (15.91)	5,921 (14.09)	0.051		23,830 (15.04)	2,834 (22.06)	0.181	
100%–199%	46,076 (21.72)	35,833 (21.94)	10,243 (20.84)	0.027		42,245 (21.51)	3,831 (24.42)	0.069	
200%–399%	103,692 (32.62)	80,403 (32.49)	23,289 (33.14)	0.014		96,741 (32.86)	6,951 (29.55)	0.071	
≥400%	114,167 (30.12)	87,972 (29.66)	26,195 (31.93)	0.049		107,084 (30.59)	7,083 (23.97)	0.149	
Health insurance coverage, *n* (%)					<.001[b]				.079[b]
Yes	280,532 (94.41)	216,614 (93.96)	63,918 (96.21)	0.104		260,487 (94.35)	20,045 (95.15)	0.036	
No	10,067 (5.59)	8,337 (6.04)	1,730 (3.79)	0.104		9,413 (5.65)	654 (4.85)	0.036	
Household smoking, *n* (%)					<.001[b]				<.001[b]
Yes	36,736 (13.69)	27,834 (13.38)	8,902 (14.91)	0.044		33,351 (13.37)	3,385 (17.86)	0.124	
No	253,863 (86.31)	197,117 (86.62)	56,746 (85.09)	0.044		236,549 (86.63)	17,314 (82.14)	0.124	
Age of mother at child's birth, *n* (%)					.343[b]				<.001[b]
18–24	45,516 (20.18)	35,007 (20.27)	10,509 (19.83)	0.011		41,217 (19.78)	4,299 (25.43)	0.135	
25–29	74,629 (25.98)	57,541 (25.84)	17,088 (26.52)	0.015		69,060 (25.89)	5,569 (27.13)	0.028	
30–34	95,508 (30.56)	74,415 (30.54)	21,093 (30.63)	0.002		89,449 (30.80)	6,059 (27.46)	0.074	

(table continues)

Table 1 *(continued)*

Characteristic	Overall	Allergic conditions: No allergy	Allergic conditions: Allergy	SMD	*p*	Asthma: No asthma	Asthma: Asthma	SMD	*p*
35–39	57,248 (17.88)	44,364 (17.97)	12,884 (17.54)	0.011		53,656 (18.13)	3,592 (14.73)	0.092	
≥40	17,698 (5.40)	13,624 (5.38)	4,074 (5.49)	0.005		16,518 (5.41)	1,180 (5.26)	0.007	
Low birth weight, *n* (%)					.318[b]				<.001[b]
Yes	24,523 (9.28)	18,747 (9.21)	5,776 (9.55)	0.012		22,028 (8.93)	2,495 (13.85)	0.155	
No	266,076 (90.72)	206,204 (90.79)	59,872 (90.45)	0.012		247,872 (91.07)	18,204 (86.15)	0.155	
Preterm birth, *n* (%)					<.001[b]				<.001[b]
Yes	31,751 (11.45)	23,577 (11.04)	8,174 (13.08)	0.063		28,252 (10.93)	3,499 (18.25)	0.208	
No	258,848 (88.55)	201,374 (88.96)	57,474 (86.92)	0.063		241,648 (89.07)	17,200 (81.75)	0.208	
ACE score group, *n* (%)					<.001[b]				<.001[b]
0	185,991 (60.60)	148,599 (62.53)	37,392 (52.94)	0.195		175,799 (61.89)	10,192 (44.00)	0.364	
1	57,847 (21.71)	43,557 (21.46)	14,290 (22.74)	0.031		52,997 (21.48)	4,850 (24.74)	0.078	
2	22,514 (8.60)	16,072 (7.92)	6,442 (11.30)	0.115		20,016 (8.21)	2,498 (13.64)	0.175	
3	11,140 (4.10)	7,823 (3.69)	3,317 (5.71)	0.096		9,783 (3.85)	1,357 (7.23)	0.148	
≥4	13,107 (4.99)	8,900 (4.40)	4,207 (7.31)	0.124		11,305 (4.57)	1,802 (10.39)	0.223	

Note. All statistical tests were conducted accounting for weighted analyses. SMDs were calculated as the absolute difference of weighted means (for continuous variables) or weighted proportions (for categorical variables) divided by the pooled weighted standard deviation. ACE = adverse childhood experience; CI = confidence interval; FPL = family poverty level; SMD = standardized mean difference.

[a] Design-based *t* test. [b] Rao–Scott adjusted χ^2 tests.

odds of parent-reported allergic conditions (1 ACE: *OR* = 1.21; 2 ACEs: *OR* = 1.59; 3 ACEs: *OR* = 1.68; ≥4 ACEs: *OR* = 1.85; $p < .001$). When the ACE score was modeled as a continuous variable, each 1-point increase was associated with 15% higher odds of allergic conditions (*OR* = 1.15; 95% CI [1.13, 1.17]; $p < .001$). Similar dose–response associations were observed for asthma. Compared with children without ACE exposure, those with ≥4 ACEs had more than double the odds of asthma (1 ACE: *OR* = 1.27; 2 ACEs: *OR* = 1.63; 3 ACEs: *OR* = 1.77; ≥4 ACEs: *OR* = 2.08; $p < .001$). When the ACE score was modeled as a continuous variable, each 1-point increase was associated with 16% higher odds of asthma (*OR* = 1.16; 95% CI [1.13, 1.19]; $p < .001$).

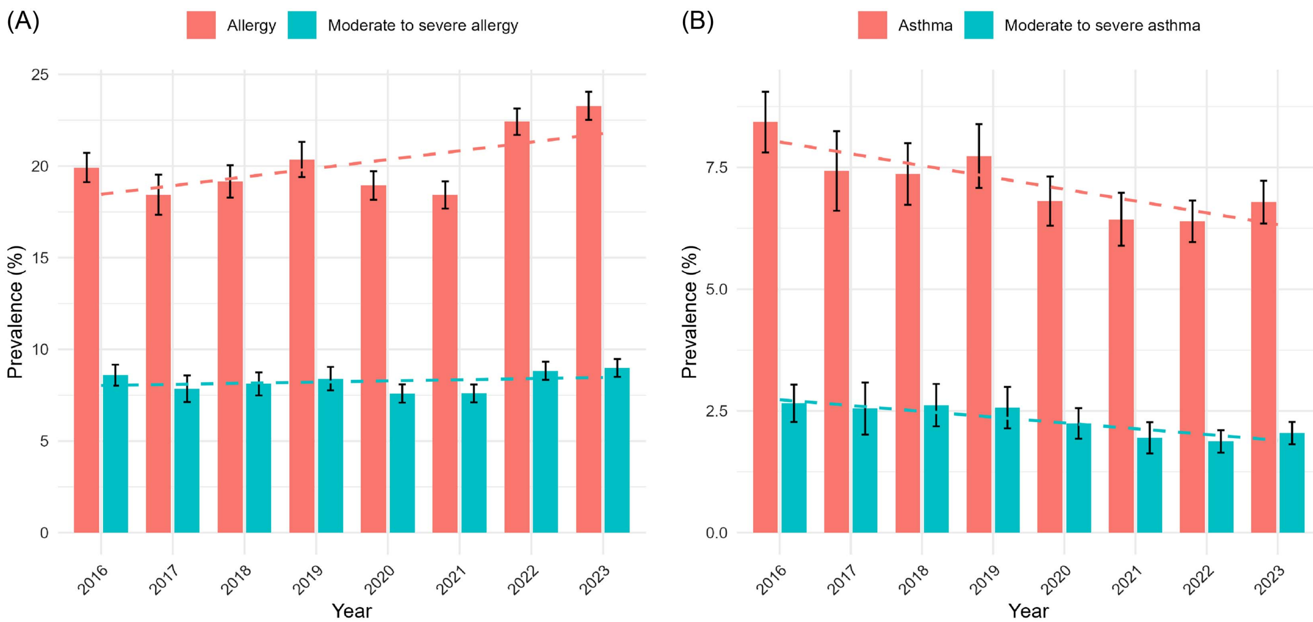
We further examined associations between individual ACE domains and parent-reported allergic outcomes (Supplemental Tables 4 and 5). After full adjustment, most individual ACE domains remained significantly associated with both parent-reported allergic conditions and parent-reported asthma. For allergic conditions, compared with children who had not experienced each respective ACE exposure, those who experienced unfair treatment due to race had higher odds of allergic conditions (*OR* = 1.62), as did those who lived with someone with mental illness (*OR* = 1.58). For asthma, compared with children who had not experienced the corresponding ACE exposure, those who lived with someone with mental illness (*OR* = 1.73), those whose families had difficulty covering basic needs such as food or housing (*OR* = 1.55), and those who experienced unfair treatment due to race (*OR* = 1.50) also had higher odds of parent-reported asthma. Notably, experiencing the death of a parent or guardian was not significantly associated with either allergic conditions or asthma after adjustments.

Associations Between ACEs and Parent-Reported Moderate-to-Severe Allergic Outcomes

When analyses were restricted to parent-reported moderate-to-severe allergic outcomes, the observed associations remained significant and appeared more pronounced (Table 3). When the ACE score was modeled as a continuous variable, each additional ACE point was associated with 17% higher odds of moderate-to-severe allergic conditions (*OR* = 1.17; 95% CI [1.15, 1.20]; $p < .001$) and 19% higher odds of moderate-to-severe asthma (*OR* = 1.19; 95% CI [1.15, 1.24]; $p < .001$). When the ACE score was treated as a categorical variable, the *OR*s for moderate-to-severe allergic conditions increased stepwise from 1.34 for 1 ACE to 2.12 for ≥4 ACEs, compared with children with 0 ACEs. For asthma, the pattern was even more pronounced: *OR*s ranged from 1.40 for 1 ACE to 2.43 for ≥4 ACEs, relative to those with 0 ACEs.

We further evaluated individual ACE domains in relation to parent-reported moderate-to-severe allergic conditions and asthma (Supplemental Tables 6 and 7). After full adjustment, most individual ACE domains remained positively associated

Figure 2
Annual Prevalence Trends of Allergic Outcomes, 2016–2023



Note. (A) Allergic conditions and moderate-to-severe allergic conditions. (B) Asthma and moderate-to-severe asthma. Estimates are weighted to account for the complex survey design. See the online article for the color version of this figure.

with both outcomes. The strongest associations with moderate-to-severe allergic conditions were observed for unfair treatment due to race (*OR* = 1.81), difficulty covering basic needs such as food or housing (*OR* = 1.69), and living with someone with mental illness (*OR* = 1.68). For moderate-to-severe asthma, the strongest associations were observed for difficulty covering basic needs such as food or housing (*OR* = 1.78), living with someone with mental illness (*OR* = 1.77), and unfair treatment due to race (*OR* = 1.75). As in the primary analyses, death of parent or guardian was not significantly associated with either

Table 2
Logistic Regression Analysis of the Association Between ACE Scores and Allergic Conditions and Asthma

Variable	Model 1[a]: *OR* [95% CI]	Model 2[b]: *OR* [95% CI]	Model 3[c]: *OR* [95% CI]	Model 4[d]: *OR* [95% CI]
Allergic conditions				
ACE score (continuous)	1.172 [1.156, 1.189]	1.119 [1.103, 1.136]	1.145 [1.126, 1.165]	1.146 [1.126, 1.166]
p	<.001	<.001	<.001	<.001
ACE score group				
0	Reference	Reference	Reference	Reference
1	1.252 [1.194, 1.312]	1.149 [1.095, 1.206]	1.209 [1.148, 1.273]	1.212 [1.151, 1.276]
2	1.685 [1.576, 1.801]	1.473 [1.376, 1.577]	1.593 [1.479, 1.717]	1.593 [1.478, 1.716]
3	1.826 [1.667, 1.999]	1.531 [1.395, 1.680]	1.665 [1.506, 1.841]	1.680 [1.519, 1.859]
4–9	1.962 [1.805, 2.133]	1.607 [1.474, 1.752]	1.836 [1.668, 2.021]	1.847 [1.675, 2.036]
p	<.001	<.001	<.001	< .001
Asthma				
ACE score (continuous)	1.279 [1.255, 1.303]	1.200 [1.175, 1.225]	1.168 [1.140, 1.197]	1.159 [1.130, 1.189]
p	<.001	<.001	<.001	<.001
ACE score group				
0	Reference	Reference	Reference	Reference
1	1.620 [1.499, 1.750]	1.362 [1.256, 1.477]	1.283 [1.176, 1.399]	1.269 [1.163, 1.384]
2	2.336 [2.110, 2.585]	1.839 [1.655, 2.043]	1.667 [1.490, 1.866]	1.629 [1.453, 1.825]
3	2.639 [2.313, 3.011]	1.995 [1.741, 2.285]	1.797 [1.558, 2.073]	1.770 [1.530, 2.048]
4–9	3.197 [2.827, 3.615]	2.382 [2.096, 2.707]	2.160 [1.867, 2.498]	2.077 [1.791, 2.409]
p	<.001	<.001	<.001	<.001

Note. ACE = adverse childhood experience; CI = confidence interval.
[a] Model 1 was adjusted for no covariates. [b] Model 2 was adjusted for age, sex, and race/ethnicity. [c] Model 3 was adjusted for age, sex, race/ethnicity, region, family structure, family poverty level, parental education, health insurance coverage, and survey year. [d] Model 4 was adjusted for age, sex, race/ethnicity, region, family structure, family poverty level, parental education, health insurance coverage, survey year, maternal age at childbirth, household smoking exposure, low birth weight, and preterm birth.

Table 3
Logistic Regression Analysis of the Association Between ACE Scores and Moderate-to-Severe Allergic Conditions and Asthma

Variable	Model 1[a]: *OR* [95% CI]	Model 2[b]: *OR* [95% CI]	Model 3[c]: *OR* [95% CI]	Model 4[d]: *OR* [95% CI]
Moderate-to-severe allergic conditions				
ACE score (continuous)	1.206 [1.185, 1.227]	1.156 [1.134, 1.177]	1.169 [1.144, 1.194]	1.171 [1.146, 1.197]
p	<.001	<.001	<.001	<.001
ACE score group				
0	Reference	Reference	Reference	Reference
1	1.412 [1.319, 1.512]	1.290 [1.203, 1.382]	1.338 [1.246, 1.438]	1.342 [1.249, 1.443]
2	1.898 [1.738, 2.074]	1.655 [1.512, 1.810]	1.742 [1.580, 1.921]	1.745 [1.583, 1.924]
3	2.003 [1.778, 2.257]	1.688 [1.497, 1.902]	1.790 [1.576, 2.032]	1.813 [1.594, 2.064]
4–9	2.300 [2.054, 2.577]	1.903 [1.696, 2.136]	2.087 [1.844, 2.363]	2.117 [1.867, 2.400]
p	<.001	<.001	<.001	<.001
Moderate-to-severe asthma				
ACE score (continuous)	1.339 [1.299, 1.380]	1.268 [1.225, 1.312]	1.205 [1.159, 1.252]	1.193 [1.147, 1.240]
p	<.001	<.001	<.001	<.001
ACE score group				
0	Reference	Reference	Reference	Reference
1	1.931 [1.684, 2.216]	1.604 [1.389, 1.854]	1.421 [1.224, 1.650]	1.402 [1.206, 1.630]
2	2.801 [2.309, 3.399]	2.190 [1.796, 2.671]	1.794 [1.476, 2.180]	1.744 [1.432, 2.123]
3	3.626 [2.860, 4.597]	2.776 [2.178, 3.540]	2.250 [1.749, 2.894]	2.203 [1.702, 2.850]
4–9	4.243 [3.460, 5.203]	3.226 [2.604, 3.998]	2.561 [2.036, 3.220]	2.430 [1.937, 3.048]
p	<.001	<.001	<.001	<.001

Note. ACE = adverse childhood experience; CI = confidence interval.
[a] Model 1 was adjusted for no covariates. [b] Model 2 was adjusted for age, sex, and race/ethnicity. [c] Model 3 was adjusted for age, sex, race/ethnicity, region, family structure, family poverty level, parental education, health insurance coverage, and survey year. [d] Model 4 was adjusted for age, sex, race/ethnicity, region, family structure, family poverty level, parental education, health insurance coverage, survey year, maternal age at childbirth, household smoking exposure, low birth weight, and preterm birth.

moderate-to-severe allergic conditions or moderate-to-severe asthma after full adjustment.

Dose–Response Associations by RCS Analysis

RCS analyses were conducted to explore potential nonlinear dose–response relationships between ACE exposure and all four outcomes, with adjustment for all covariates in Model 4. For allergic conditions (Figure 3A), the adjusted odds increased steadily, with a slower rate of increase beyond 5 ACEs. A similar trend was observed for asthma (Figure 3B). Notably, the dose–response associations were even more pronounced for parent-reported moderate-to-severe allergic conditions (Figure 3C) and parent-reported moderate-to-severe asthma (Figure 3D), where risks began to rise more sharply beyond 3 ACEs and continued increasing through 8–9 ACEs. All associations were statistically significant and showed strong evidence of nonlinearity (all p-overall <.001; all p-nonlinear <.001).

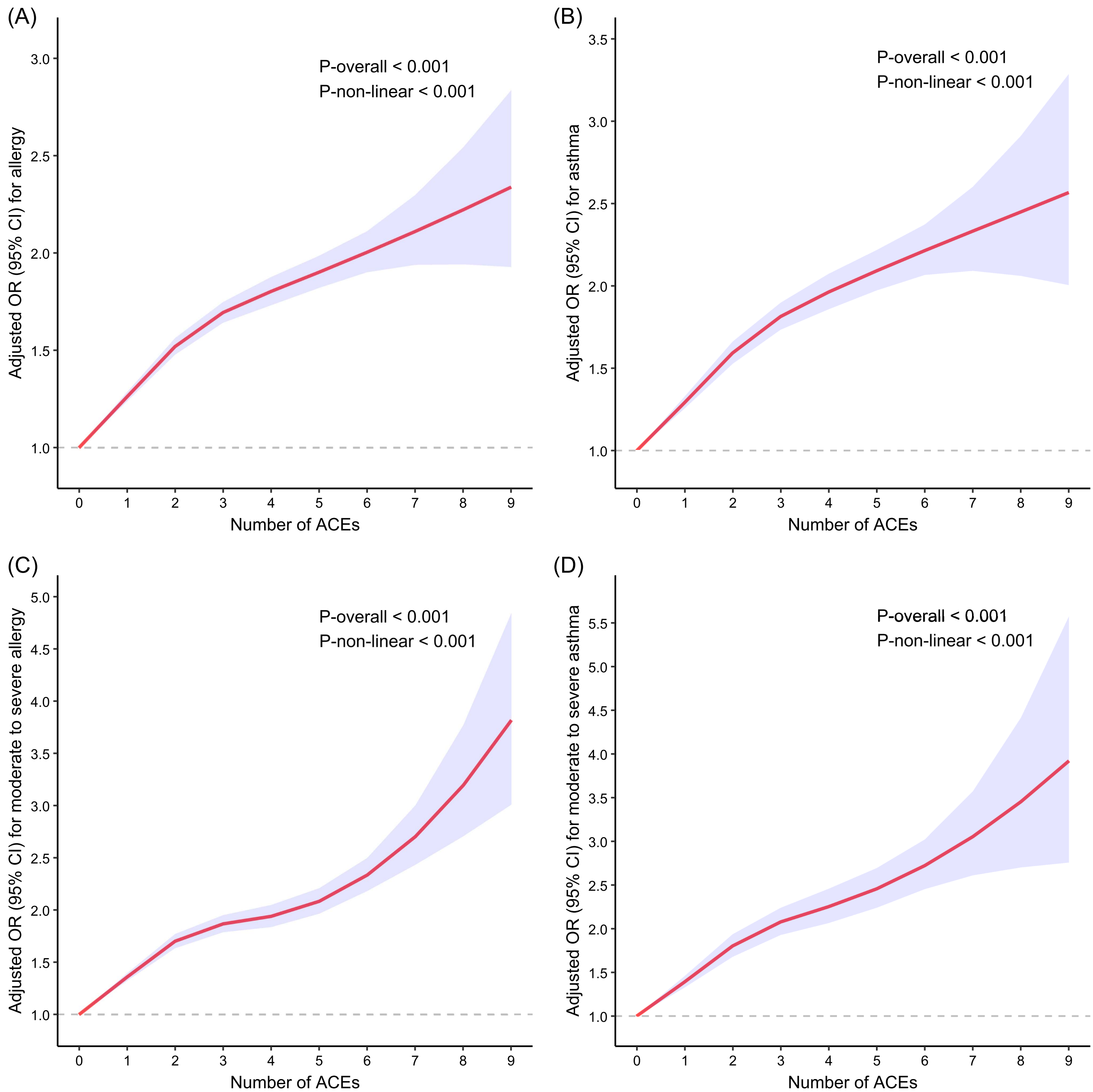
Subgroup and Interaction Analyses

Subgroup analyses showed broadly consistent associations across demographic and household factors, including age, sex, race/ethnicity, FPL, and household smoking exposure (Figure 4). Among these, age demonstrated the clearest effect modification (p for interaction <.001 for allergic conditions and .002 for asthma). Compared with children aged 0–2 years with 0 ACEs, those with ≥4 ACEs had higher odds of parent-reported allergic conditions (*OR* = 2.64); the corresponding association in adolescents aged 12–17 years was weaker (≥4 ACEs vs. 0 ACEs: *OR* = 1.71). A similar dose–response gradient was observed for asthma, with the strongest association in children aged 0–2 years: Compared with those with 0 ACEs, children with ≥4 ACEs had higher odds of parent-reported asthma (*OR* = 4.71). The corresponding association in adolescents aged 12–17 years was weaker (≥4 ACEs vs. 0 ACEs: *OR* = 2.02). Age-stratified weighted prevalences of allergic conditions and asthma, and the weighted distribution of ACE score groups, are presented in Supplemental Figure 3 and Supplemental Table 8.

FPL also modified the association between ACEs and allergic conditions (p for interaction = 0.031). Among children living in families with <100% of the FPL, the odds of allergic conditions increased substantially with increasing ACE exposure (1 ACE: *OR* = 1.26; 2 ACEs: *OR* = 1.70; 3 ACEs: *OR* = 1.78; ≥4 ACEs: *OR* = 2.16), compared with those with 0 ACEs. In contrast, children in families with ≥400% of the FPL also had elevated odds of allergic conditions (1 ACE: *OR* = 1.13; 2 ACEs: *OR* = 1.38; 3 ACEs: *OR* = 1.34; ≥4 ACEs: *OR* = 1.49), but these odds were consistently lower than those observed in the <100% FPL group. For asthma, although numerically higher ORs were observed in some lower FPL groups, the interaction was not statistically significant (p for interaction = 0.879).

Additional subgroup analyses (by family structure, region, and ACEs modeled as a continuous variable) were presented in Supplemental Figures 4–7 to maintain clarity and visual readability in the main figure.

Figure 3
Restricted Cubic Spline Curves Illustrating the Association Between ACE Scores and Allergic Outcomes



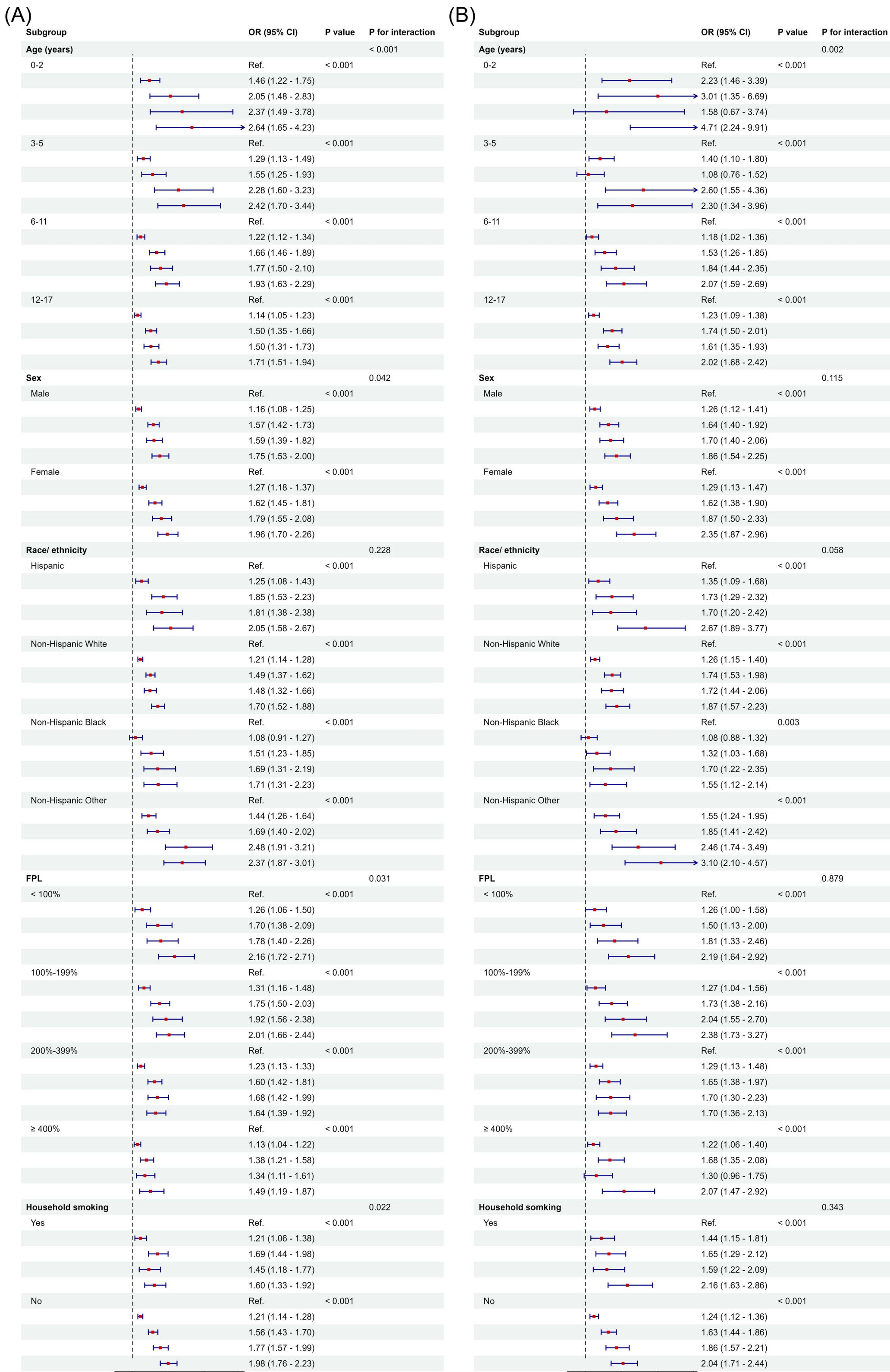
Note. The figure shows the following: (A) allergic conditions, (B) asthma, (C) moderate-to-severe allergic conditions, and (D) moderate-to-severe asthma. Analyses were based on Model 4, adjusted for age, sex, race/ethnicity, region, family structure, family poverty level, parental education, health insurance coverage, survey year, maternal age at childbirth, household smoking exposure, low birth weight, and preterm birth. The red line represents the odds ratio, and the shaded area represents the 95% confidence interval. ACE = adverse childhood experience; CI = confidence interval. See the online article for the color version of this figure.

Discussion

Psychological stress and early adversity are increasingly recognized as key determinants of both mental and physical health in childhood (Moriya et al., 2024; Russell et al., 2025; Weems et al., 2021). Despite the significant burden of allergic diseases (Koo et al., 2023; Shin et al., 2023) and their bidirectional links with psychological well-being (Xu et al., 2025), the role of early-life psychosocial adversity, particularly ACEs, in the development of allergic conditions remains insufficiently understood. In this large, nationally representative sample of 290,599 U.S. children, we observed robust,

Figure 4

Subgroup Analyses of Associations Between Adverse Childhood Experience Score Groups and Allergic Outcomes, Stratified by Age, Sex, Race/Ethnicity, FPL, and Household Smoking Exposure



Note. The figure shows the following: (A) allergic conditions and (B) asthma. Models were adjusted for all covariates in Model 4 except the stratification variable. The vertical dashed line represents an odds ratio of 1. CI = confidence interval; FPL = family poverty level; Ref. = reference. See the online article for the color version of this figure.

consistent, and dose-dependent associations between cumulative ACE exposure and both allergic conditions and asthma. These associations between ACE exposure and parent-reported allergic conditions and asthma were especially pronounced in younger children and, for allergic conditions, among children living in families with lower FPL, highlighting ACEs as a potential social determinant of pediatric allergic disease and an important target for prevention.

Following Census Bureau recommendations for multiyear analyses, we pooled data from the 2016–2023 NSCH and applied survey weights to construct a nationally representative analytic sample. During this period, the weighted prevalence of allergic conditions increased from 19.94% in 2016 to 23.28% in 2023, consistent with prior epidemiological reports (Pate et al., 2022). More importantly, ACEs were strongly and dose-dependently associated with allergic conditions. Compared with children with no ACEs, those with 1 ACE had 21% higher odds of parent-reported allergic conditions, whereas those with ≥4 ACEs had nearly twice the odds (*OR* 1.85; 95% CI [1.68, 2.04]), even after full adjustment. These findings align with the broader literature linking ACEs to adverse pediatric outcomes—including anxiety, depression (Matthews et al., 2024), neurodevelopmental disorders (Walker et al., 2022), and headaches (Mansuri et al., 2020)—and extend this evidence to allergic disease risk, suggesting that allergic conditions may also constitute part of this broader spectrum of ACE-related pediatric health outcomes.

Our findings also confirmed strong, dose-dependent associations between ACEs and asthma. Each additional ACE was associated with 16% higher odds of parent-reported asthma, and children with ≥4 ACEs had more than twice the odds of parent-reported asthma compared with those with no ACEs. These findings build on earlier national and regional studies by incorporating more recent NSCH data and applying a harmonized analytic framework (Exley et al., 2015; Ross et al., 2021; Wing et al., 2015). While our focus was on childhood exposures, prior research suggests that stress-related risk for asthma may begin accumulating much earlier. Maternal experiences of anxiety, depression, major negative life events, or exposure to violence during pregnancy have been linked to higher risks of wheeze and asthma in offspring (Adgent et al., 2024; Osam et al., 2023). Importantly, many adversities captured within the ACE framework are experienced primarily by caregivers, particularly mothers, rather than by children alone. These persistent, family-level stressors may serve as key pathways linking ACEs to asthma. Our findings underscore the importance of considering both maternal and child exposures when interpreting the impact of ACEs across development.

Cumulative Burden and Heterogeneity of ACEs

Since it was first proposed by Felitti et al. (1998), the ACE framework has emphasized the cumulative burden of early-life adversities, offering a polyvictimization model to quantify stress exposure across multiple domains. In our study, we observed a clear, dose-dependent association between ACEs and both allergic conditions and asthma, confirmed by RCS models. Notably, children with ≥4 ACEs had significantly higher odds of allergic conditions and asthma, aligning with previous research suggesting a threshold effect, where the risk escalates sharply with exposure to multiple adversities (Hamby et al., 2021). Existing models, including the resilience portfolio model, underscore the cumulative and nonlinear interactions among diverse adversities, illustrating how the cumulative burden of trauma shapes health outcomes (Banyard et al., 2025). Beyond the ACE structure, families experiencing multiple ACEs often face additional structural disadvantages, such as poor housing quality, limited access to health care, and heightened maternal stress, creating a vicious cycle that amplifies health risks. This interplay reinforces vicious cycles of adversity, in which the accumulation of stressors further exacerbates negative health outcomes (Hamby et al., 2021). Our study indicates that this cumulative burden and such reinforcing cycles of disadvantage may also shape the relationship between ACE exposure and allergic conditions.

However, when examining individual ACEs, not all exposures conferred equivalent risk for allergic conditions and asthma. Most individual ACE domains remained positively associated with allergic outcomes after full adjustment, and the pattern was broadly similar for the corresponding moderate-to-severe outcomes. In particular, in our study, caregiver mental illness and experiences of discrimination showed the strongest associations with both allergic conditions and asthma. A large U.K. primary care cohort similarly reported increased risks of asthma and allergic rhinitis in children exposed to common maternal mental illness, such as depression and anxiety (Osam et al., 2023). Likewise, in a multicenter study of adolescents from racial and ethnic minority backgrounds, perceived discrimination was linked to both higher asthma prevalence and worse asthma control (Thakur et al., 2017). It is possible that chronic adversities such as caregiver mental illness exert more sustained biological and environmental influences than acute, one-time shocks like bereavement, whose long-term impact may depend on buffering resources and recovery context (Lucente & Guidi, 2023; Osam et al., 2023). In contrast, caregiver death exhibited weaker and more inconsistent associations in our analysis, with no significant associations remaining after covariate adjustment. Although bereavement can be an intensely distressing experience for children and may influence long-term health outcomes, its downstream effects may depend more heavily on the availability of stable caregiving and a supportive environment (Liu et al., 2022).

Although precise mechanisms remain under investigation, chronic or repeated adversity may be biologically embedded via persistent activation of stress-response systems. Chronic activation of the hypothalamic–pituitary–adrenal axis and autonomic nervous system may alter immune development and regulatory capacity (Engel et al., 2021; Soares et al., 2021). These changes can disrupt immune homeostasis, biasing immune profiles toward Th2 polarization and reducing tolerance (Reid et al., 2024; Taves & Ashwell, 2021). This, in turn, may lower immune tolerance and increase susceptibility to allergic sensitization (Crooke et al., 2019; Oh et al., 2019). While these biological changes offer one explanatory pathway, they likely interact with psychological and social processes to shape allergic outcomes.

Developmental and Socioeconomic Vulnerability

Younger children may be more vulnerable to the impact of ACEs, with the strongest associations observed between ACEs and both allergic conditions and asthma in children aged 0–2. Notably, although the baseline prevalence of allergic conditions and asthma was lower among younger children, the associations between ACE exposure and parent-reported allergic conditions or asthma were stronger in early childhood. Early childhood is marked by immune and microbial maturation, characterized by Th2-skewed adaptive immunity, limited memory T/B cell reservoirs, and heightened sensitivity to environmental programming (Dowling & Levy, 2014; Ygberg & Nilsson, 2012). During this period, adversity-induced epigenetic and neuroendocrine changes may become biologically embedded, resulting in long-term alterations in immune regulation and inflammatory tone (Chen & Lacey, 2018; Soares et al., 2021). Concurrently, the infant gut microbiome undergoes rapid colonization, and its composition plays a crucial role in shaping immune tolerance and regulating inflammatory responses (Donald & Finlay, 2023; Robertson et al., 2019). Disruption of this process by adverse exposures may deplete beneficial taxa and promote proinflammatory species, amplifying respiratory inflammation and allergic susceptibility through the gut–lung axis (Robertson et al., 2019). Importantly, greater exposure to the home environment may also partly contribute to this pattern, as younger children spend more time in the household setting and may therefore be more affected by indoor allergens, caregiver emotional distress, and other family-level adversities (Osam et al., 2023; Perry et al., 2024). Taken together, these converging mechanisms may explain the heightened vulnerability of younger children to the immunological consequences of early-life adversity.

Low-income households have been consistently associated with poorer mental and physical health in childhood (Jones et al., 2025; Wilson et al., 2025). In this study, the association between ACEs and allergic conditions was stronger among children from lower income households (FPL $< 100\%$, ≥ 4 ACEs, $OR = 2.16$) than among those from higher income households (FPL $\geq 400\%$, ≥ 4 ACEs, $OR = 1.49$). Children in low-income households are more likely to live in poorly maintained housing and experience greater exposure to indoor allergens such as mold, cockroaches, and rodents, increasing their vulnerability to allergic sensitization and asthma (Perry et al., 2024). In addition, allergen-free foods are often substantially more expensive, posing further challenges for allergy management in resource-limited settings (Tseng & Vastardi, 2023). Notably, although no statistically significant interaction was detected, stratified estimates suggested that the association between ACEs and parent-reported asthma remained evident across income levels and was numerically stronger in some lower FPL strata. Indeed, prior studies have consistently linked low income with higher asthma prevalence and increased emergency department utilization (Grant et al., 2022). Future studies incorporating more detailed environmental, behavioral, and health care–related measures are needed to better disentangle these context-specific mechanisms.

Implications for Practice

Allergic conditions, including asthma, may represent part of the broader spectrum of ACE-related health outcomes. These conditions may reflect not only biological vulnerability but also physiological consequences of early-life psychosocial stress. Although ACE screening often focuses on behavioral or metabolic risks, the present findings support incorporating allergic conditions into risk identification frameworks and suggest that family stress and broader social contexts may contribute to poor symptom control and recurrent exacerbations.

Simple, noninvasive questions about a child's environment, such as caregiver distress, family conflict, experiences of discrimination, or major life disruptions, may help identify children who are at heightened vulnerability to allergic conditions and asthma. For children with frequent emergency visits, medical management may be complemented by behavioral and emotional support, parenting interventions, or referral to supportive services. Associations between ACEs and allergic outcomes were stronger among children from lower income families and those living in less traditional household structures, where housing conditions, access to health care, and caregiving stability may be more compromised. Caregiver mental health problems showed one of the strongest associations with parent-reported allergic conditions and asthma in item-level analyses, suggesting that caregiver functioning may be relevant to effective disease management. Experiences of unfair treatment were also associated with higher odds of parent-reported allergic conditions and asthma, underscoring the importance of sensitive communication and timely referral to resources. At the policy level, expanding access to caregiver mental health services, family support programs, and antidiscrimination initiatives may help reduce the compounded burden and improve outcomes for allergic diseases.

Limitations and Future Directions

Several limitations should be acknowledged. First, the cross-sectional nature of the NSCH data precludes causal inference, and the temporality of associations between ACEs and allergic outcomes cannot be firmly established. Second, both ACE exposures and allergic outcomes, including disease severity, were reported by parents without external validation. This single-source, single-method approach may introduce recall or reporting bias, and common method variance could have inflated some associations. Caregiver distress or perception may influence the reporting of both adversity and illness. However, the observed associations were not uniform across individual ACE domains. This heterogeneity suggests that common method variance may not fully account for the overall pattern of findings, although it remains an important potential source of bias. Moreover, the use of brief, caregiver-rated severity items (mild/moderate/severe) may be subject to subjective interpretation and misclassification, particularly for conditions such as food allergy, where nonallergic sensitivities or intolerances may be mistaken for IgE-mediated allergy. In addition, although these items were based on prior clinician diagnoses, variation in respondents' understanding of terms such as "asthma" or "allergies" may still have introduced some measurement variability. Third, due to constraints of the NSCH data, specific subtypes such as atopic dermatitis or allergic rhinitis could not be separately analyzed. Fourth, parent-reported ACEs may diverge from children's actual experiences, especially regarding family-related adversities. Finally, although we adjusted for a wide range of demographic and socioeconomic covariates, residual confounding cannot be ruled out, particularly from unmeasured genetic or environmental factors.

Future studies should adopt prospective designs to clarify the temporal relationship between childhood ACEs and allergic outcomes and to examine whether these associations persist across developmental stages. Combining parent-reported data with clinical information, such as physician diagnosis or electronic health records, could improve assessment accuracy. Using clearer diagnostic criteria and objective indicators may enhance outcome classification and support interpretation of disease severity. Finally, multi-informant approaches, including adolescent self-reports when developmentally appropriate, may help resolve discrepancies in reports of family adversity and provide a more comprehensive understanding of ACE exposure.

Offsetting these limitations are several strengths of the study. Leveraging a nationally representative NSCH sample of 290,599 children, representing over 61 million U.S. children aged 0–17 years, this study captures a broad pediatric population across diverse sociodemographic backgrounds. A dose-based cumulative approach was integrated with item-level ACE indicators to provide a more nuanced characterization of adversity in relation to allergic outcomes. Findings remained consistent across key subgroups and were robust to adjustment for a comprehensive set of demographic and socioeconomic factors. Together, these features strengthen the public health relevance of our findings and underscore the value of integrating adversity-informed approaches into pediatric allergy research and care.

Conclusion

In this nationally representative study, cumulative ACEs were strongly and dose-dependently associated with parent-reported allergic conditions and asthma among U.S. children, particularly in early childhood. These findings suggest that early-life adversity may represent an important social determinant of pediatric allergic and respiratory health and support early identification and trauma-informed approaches to pediatric allergy and asthma care.

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